

USING AI TO BRIEF DATACENTER REMOTE HANDS

A Picture Is Worth 1024 Words

Give them a K.

The person at the rack cannot read your mind, your silkscreen, or your inventory database. Everything you want done has to survive the trip through a noisy room to someone who has never seen this chassis before.



You are not in the room

Every instruction you send crosses a gap. The tech at the rack is often a colo contractor with varying degrees of experience— and the room is actively hostile to communication.

01

Not your team

Colo remote hands may be a shared pool. They may not have built this fleet and may not see it again tomorrow.

02

Hands are full

A drive in one hand, a carrier in the other. They are not typing, and they are not scrolling.

03

It is loud

Fan walls run 75–85 dBA. Hearing protection is PPE, not an option they can decline.

04

One shot

Pull the wrong drive from a degraded pool and you are explaining an outage, not a swap.

The failure mode is not that the tech is careless. It is that you sent an instruction only you could follow.

RULE 1

Never use voice

Voice feels faster. It is the single most expensive shortcut in this whole workflow.

TRANSCRIPTION

Wrong even in a quiet room

Letter–digit strings are worst case for speech: no linguistic context to correct against.

THE ROOM

And this is not a quiet room

Fan noise masks exactly the consonants that separate B, D, E, G, P, T and V.

PPE

They are wearing ear protection

Asking them to lift a muff is asking them to remove PPE in the loudest room you own.

HEARD OVER A PHONE, IN THE AISLE

P1-DIMMG1 ↔ P1-DIMMD1

bay 15 ↔ bay 50

slot B2 ↔ slot D2

sdb ↔ sdd

No scrollbar. No screenshot. No second read.

If it has to be said out loud, it is going to be said twice — and confirmed wrong once.

Move it to chat — Slack, Teams, whatever they have

A chat channel is the only medium that is synchronous and asynchronous at the same time, and the only one that carries pictures back.

BOTH MODES

**Sync when you are both there,
async when you are not**

They read it when they get to the cage. You answer when you wake up. Neither of you has to hold the other's schedule.

INBOUND

Photos and video come back

“Is this the one?” with a photo settles in seconds what three paragraphs cannot. Video catches an intermittent LED or a fan you can hear.

ARTIFACTS

URLs, files, scrollback

Link the manual page. Paste the grid. It is all still there at shift change, and still there when someone asks what happened.

AND IT IS A RECORD

Timestamped, searchable, and attributable — for free

You get an audit trail of exactly what was asked and what came back, without anyone writing a report. That is worth as much after the maintenance as during it: when a drive turns up in the wrong bay three weeks later, the channel says who put it there and what they were told.

Reliable connectivity is a hard requirement

Everything above assumes the tech can reach you from where the hardware is. In a colo, that assumption is frequently false — and not yours to fix.

WHAT YOU ACTUALLY GET

Provider Wi-Fi is flaky or marginal

Coverage is designed for the office, not the cold aisle. Cages are steel mesh. Access points sit at the end of the row. Signal dies exactly where the racks are — and it is worst inside the cage you are paying for.

Cellular is no better: you are in a concrete box.

AND YOU CANNOT FIX IT

It is the provider's network

You cannot add an AP, move one, or open a ticket that gets triaged this week. The escalation path runs through your account manager, not your NOC.

So plan around it: confirm the tech has a working path before they are standing at the rack, and hand them something that still works when the link drops.

This is the argument for paper, and for the printer at the end of this deck: an instruction that has already arrived cannot go offline.

“Pull slot #11” is not an instruction

Remote hands may have never worked inside a server. Your shorthand carries a mental model they were never given.

WHAT YOU SEND

`pull slot #11`

`add DIMMs in P1-C1 and P2-C1`

`reseat the riser`

`swap the drive in bay 6`

WHAT IT QUIETLY ASSUMES

that “slot” means a drive bay here, and a PCIe slot two lines later

that P1 is a CPU, that C is a channel, and that the board is labeled at all

that they know which card is the riser, and that it lifts rather than slides

that bays are numbered the way you are counting them

Every one of those assumptions is a place the job silently goes wrong — and the tech will not know to ask.

The legend is there. They still cannot read it.

Most boards and chassis do carry slot markings. Assuming a tech can find and use them is a different claim entirely.

COVERED

Cabling sits on the silkscreen

DIMM slot names are routinely hidden under power and fan harnesses.

HIDDEN

Under the air shroud

On some boards the shroud has to come off to see the markings

AMBIGUOUS

Tiny, unlit, and upside down

Silkscreen is low-contrast gray on green, 2 mm tall, in a dark rack, read from whichever side the rails let you stand on.

THE PRACTICAL CONSEQUENCE

Anchor instructions to landmarks nobody has to read

The CPU socket, the front of the chassis, the rear I/O panel, the handles. These are visible with the lid off, from across the aisle, under a headlamp — and they do not depend on the tech finding 2 mm of gray text. Use the slot name as confirmation, never as the address.

Which one is bay 6?

Same twelve bays, same front panel, four legitimate numbering conventions. Vendors are not consistent — and it varies between a 1U and a 2U of the same generation, from the same vendor.

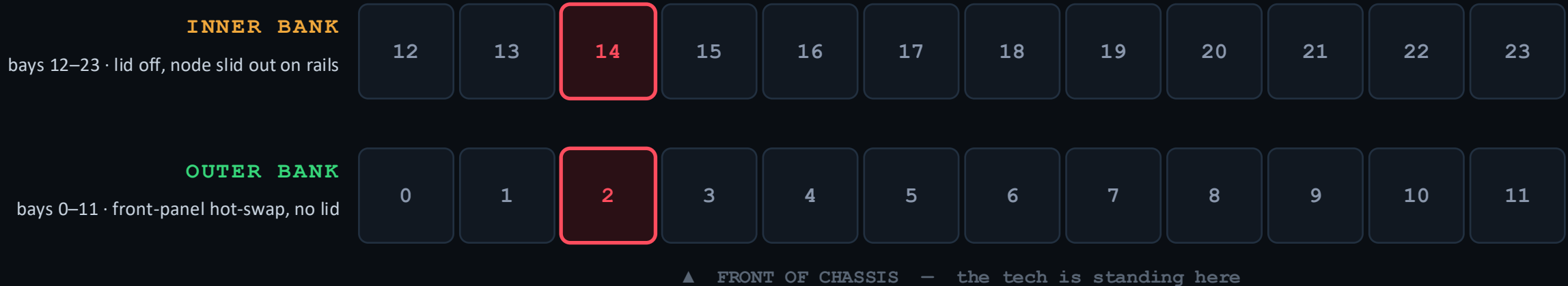
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“Bay 6” is four different drives. Origin corner, index base and major axis all have to be stated — or shown.

AND THEN THERE IS DEPTH

Inner bank or outer bank?

On a top-loading or dual-bank “missile launcher” chassis, the bay grid runs front-to-back as well as side-to-side. “Third from the left” now has two answers, and they are not equally safe to get wrong.



TWO ANSWERS

Both highlighted bays are “third from the left”

One is bay 2. The other is bay 14. The tech cannot tell which you meant from the position alone.

AND THEY ARE NOT THE SAME JOB

The inner bank is a lid-off, slide-the-node-out job

Different risk, different time, different change window. Name the bank every time — and say whether the lid comes off.

A picture is worth 1024 bytes. Give them a K.

Everything so far is the same problem: you are encoding a physical location as text, and they have to decode it standing in front of the metal. Skip the encoding. Draw it.

NO DECODING

The picture is the location

A highlighted cell in the right place needs no convention, no index base, and no origin corner. There is nothing to get backwards.

SELF-CHECKING

They can verify before they pull

Shape of the grid, position of the CPU, which end the handles are on — all of it has to match what is in front of them, or they stop and ask.

TEACHES

Slot names on the picture

Print P1-DIMMG1 on the cell anyway. It is the read-back check, and after three jobs the tech has learned the scheme for free.

This is the part AI makes cheap. Generating a correct, chassis-specific diagram used to cost more than the maintenance was worth, so nobody did it — they typed “bay 6” and hoped. A model that has read the manual can draw the right grid, for the right chassis, with the right bay lit, in the time it takes to open the ticket.

Show the board, not the slot name

Draw the DIMM slots where they physically are: relative to the front of the chassis and to the CPU sockets. Then label them.

ORIENTATION

State which way they are standing

“FRONT OF NODE” up top, “REAR — YOU STAND HERE” below.

LANDMARKS

CPU sockets drawn in place

Both sockets, in their real positions and at their real size.

SLOT NAMES

P1-DIMMG1 on the cell itself

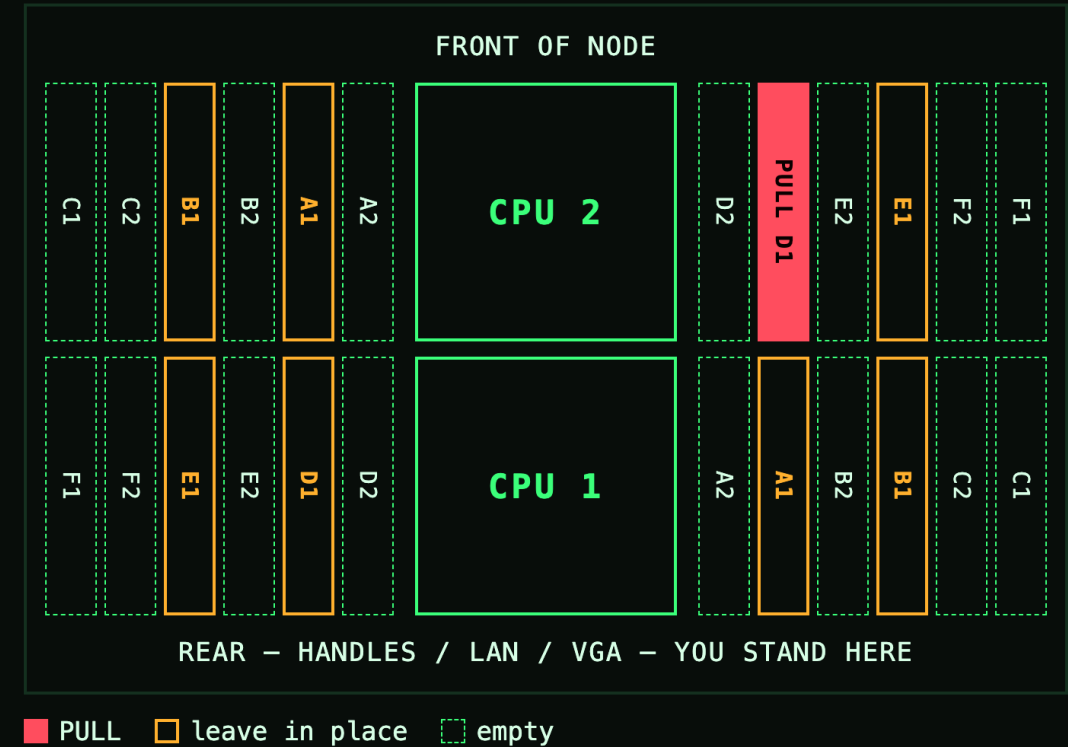
Verification now, and repetition that teaches the scheme.

One red cell. Everything else is orientation.

node17 was **srv202**

rack 02 · U17 · quad01 · bay C · X11DPT-B

PULL P2-DIMMD1 · **Samsung 64GB**



The same picture says where they go

A population plan is the same drawing with the action inverted — fill these, leave those empty. Channel rules are decisions the tech should never have to make at the rack.

ONE ACTION PER CELL

“install 32GB” or “leave empty”

No legend lookup. The cell says what happens to it.

THE RULE, STATED

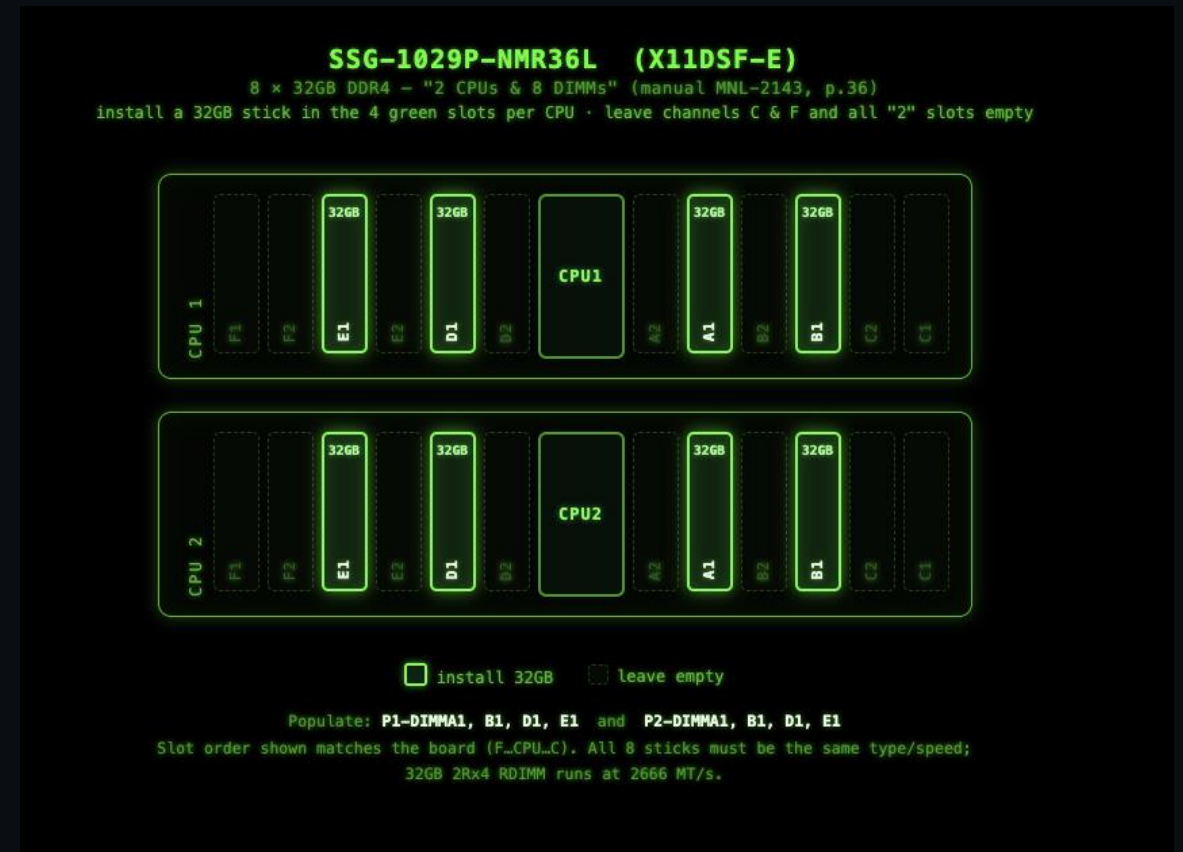
Channels C and F stay empty

The rule is in the manual. Put the outcome on the page.

CITED

Manual MNL-2143, p.36

How you check the model got it right, and how hands escalate.



Light up the bay, not the sentence

Same treatment for drives: the real grid shape, in the real orientation, with the one bay in question highlighted and everything else visibly not-it.

ONE HIGHLIGHT

One fault, one hot cell

Eleven calm cells and one red one. The eye finds it first.

POSITION IN WORDS TOO

“top-right of the front panel”

Catches the case where the picture printed in grayscale.

THE READ-BACK

Confirm the serial before pulling

The grid finds the drive; the serial on the label proves it.

node08 — SSG-6029P-E1CR24L

rack 07 · U32–U33 (2U) · FRONT bank · pull bay 11 · replace 20 TB SAS (ST20000NM002D) — head-select fault

● healthy ● pull this one bay # = SES slot

FRONT BANK (BAYS 0–11)

bay 2 ● sdc osd.103 ZVT7QH1D	bay 5 ● sdf osd.106 ZVTA5JS9	bay 8 ● sdi osd.109 ZVT5WG8B	bay 11 ● sdl PULL ZVTCBYDM
bay 1 ● sdb osd.102 ZVTB4XR7	bay 4 ● sde osd.105 ZVT6NP0W	bay 7 ● sdh osd.108 ZVTD3RF6	bay 10 ● sdk osd.111 ZVT4ZC5X
bay 0 ● sda osd.101 ZVT9M2KQ	bay 3 ● sdd osd.104 ZVTC8LM3	bay 6 ● sdg osd.107 ZVT8KD2V	bay 9 ● sdj osd.110 ZVTE7YN4

PULL bay 11 — top-right of the FRONT panel (amber locate LED is lit). Confirm serial **ZVTCBYDM** on the drive label before pulling. Replace with any 20 TB SAS 3.5" drive in the same bay.

Red means scrap

A single fault is the easy case. A real job has two actions at once — and here the difference between them is fourteen drives going in the bin and twenty-six good ones going back into service.

RESERVE THE RED

Red is “this part is bad” — nothing else

Amber is “healthy, and moving”. Color those the same and twenty-six working 15 TB drives go in the scrap bin.

ENUMERATE EVERY PULL

By bay. Never by count, never by model

Hands cannot see what is in a bay before they pull it, so “pull any 8” is not executable. Eight bay numbers are.

NF1 pull & relocate — ceph4 consolidation

All drives Samsung MZ4LB15THMLA (15.36 TB). Anonymised: hostnames, chassis serials, rack and RU removed. Drive serials retained.

NODE-1 SSG-1029P-NMR36L · X11DSF-E · BIOS 3.1 21 of 32 bays populated · PULL 8 · MOVE 13	NODE-2 SSG-1029P-NMR36L · X11DSF-E · BIOS 4.7 19 of 32 bays populated · PULL 6 · MOVE 13
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■ PULL — scrap, do not reuse ■ MOVE — healthy, relocate to d02/d03/d05/d08 □ empty bay

NODE-1					
bays 0–15			bays 16–31		
0	● PULL	S4DENY0M600619	16	MOVE	S4DENY0M600459
1	MOVE	S4DENY0M600355	17	MOVE	S4DENY0M600758
2	MOVE	S4DENY0M600617	18	—	empty
3	—	empty	19	—	empty
4	● PULL	S4DENY0M600388	20	MOVE	S4DENY0M600143
5	● PULL	S4DENY0M600377	21	MOVE	S4DENY0M600467
6	MOVE	S4DENY0M600590	22	MOVE	S4DENY0M600473
7	—	empty	23	—	empty
8	MOVE	S4DENY0M600139	24	MOVE	S4DENY0K900333
9	MOVE	S4DENY0M600562	25	● PULL	S4DENY0M600142
10	MOVE	S4DENY0M600471	26	● PULL	S4DENY0M600419
11	—	empty	27	—	empty
12	● PULL	S4DENY0M600413	28	● PULL	S4DENY0K900342
13	MOVE	S4DENY0M600439	29	—	empty
14	● PULL	S4DENY0M600378	30	—	empty
15	—	empty	31	—	empty

One of two nodes from a real consolidation. Serials kept so the tech can confirm; hostnames, chassis serials, rack and RU stripped before it left the building.

BEFORE ANY OF THAT

Make sure they are at the right chassis

A perfect bay diagram applied to the wrong server is worse than no diagram at all — now you have an outage and a confident tech.

NEVER TRUST THE STICKER

Adhesive labels go stale after a rename and fall off in a hot aisle — and the failure mode is not that the label is missing. It is that someone found it on the floor and stuck it back on the wrong chassis.

A label is a hint. It is never the identity.

USE THIS

Locator LED

Light it from your side, then ask what is blinking. It is the only identifier that cannot be moved by hand.

AND THIS

Chassis serial number

Stamped or etched by the vendor, and the thing your inventory actually keys on. Have them read it back.

Lighting that LED is not one command. It varies by vendor, by system model, and even across BIOS and BMC revisions on the same model — sysfs, ipmitool, redfish, or a vendor tool, each with its own spelling. This is exactly the kind of fiddly, well-documented, per-model lookup to hand to the AI: let it work out how to light this box, rather than maintaining the matrix yourself.

Distill the sources into a skill

None of this works if the model is guessing at geometry. Feed it the documentation once, verify it, and it stops guessing.

SOURCE 01

Vendor manuals

User, maintenance and technical guides for every chassis and board in the fleet. The full manual, not the quick-reference guide — the QRG omits the low-count population rows.

SOURCE 02

Datasheets and spec sheets

Bay counts, backplane topology, DIMM channel maps, supported ranks and speeds, riser options.

SOURCE 03

Labeled photos

Vendor product shots, teardowns, review-site photography. A color photo of a bare board settles orientation questions a figure cannot.

SOURCE 04

What you learn in the field

Every time hands correct you, that is a fact the manual did not have. Write it back into the skill.

Then verify before you trust it. Manual figures mirror silently — two label lists running in opposite directions are both self-consistent, so a wrong reading looks right. Confirm geometry against a live host (dmidecode, SES slot enumeration) or a photograph of the real board, and pin the result. Derive it once, carefully; reuse it forever.

Know what works for your tech

The best card is the one this person will actually use. That is not a property of the card — it is a property of the pairing, and you only learn it by watching.

SOME WANT THE PICTURE

Rich graphics, color, board geometry

Reads spatially. Finds the slot from the shape of the drawing and never looks at the legend. For this tech the diagram is the whole instruction — more words make it worse.

OTHERS WANT THE LIST

Plain monospace text, 60 columns, no color

Bounced off the graphical card entirely. Wants a bordered text grid per host: draw lines, return lines, a two-column slot/action table, one summary line. Same data, no graphics.

Both of those are real preferences from real technicians, and the second only surfaced because a card came back unused. Ask, or infer it from past friction with that specific person — and keep the generator able to emit both formats from one dataset, so switching costs nothing. Do not assume the prettier artifact is the better one.

Put an all-in-one printer in the datacenter

Even if the techs have to reach it over Bluetooth or the printer's own Wi-Fi. This is the cheapest item in the whole workflow and the one people leave out.

HANDS FULL

Paper is easier to carry than a phone

They are holding a drive and a carrier. Paper goes in a pocket, under an arm, or on top of the chassis — and it does not lock, sleep, or need a clean finger.

NO SCROLLING

The whole job is visible at once

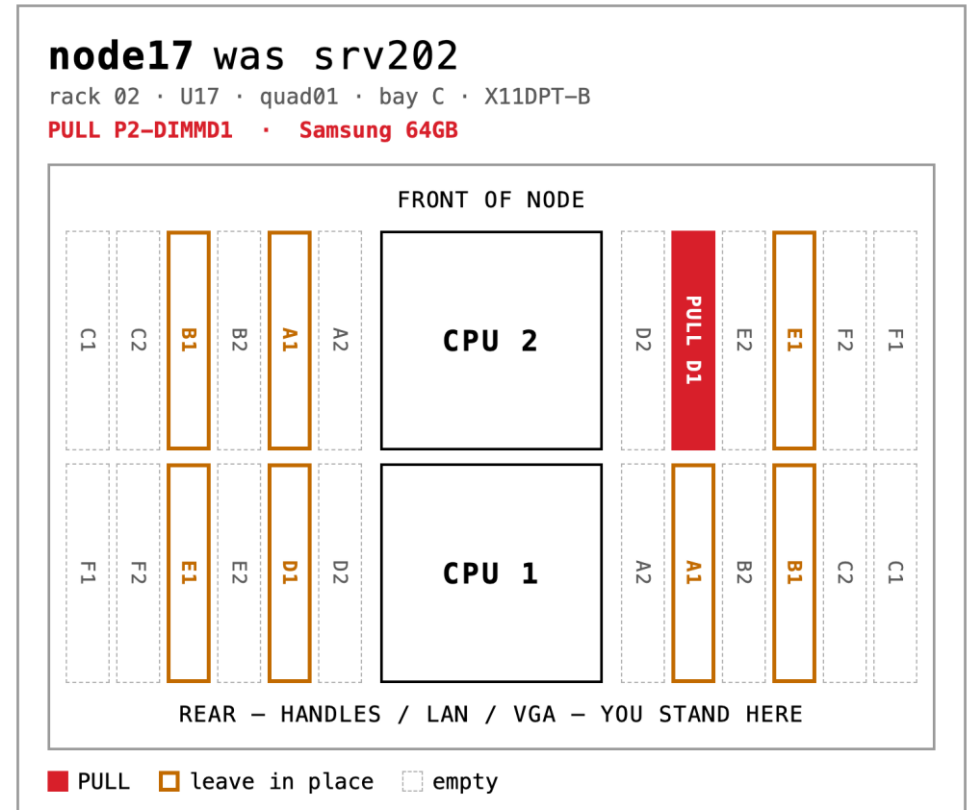
Few people have an $8\frac{1}{2} \times 11$ phone. On paper the grid, the slot list and the confirmation note are in view together, which is exactly when a tech catches a mismatch.

OFFLINE BY DEFAULT

It already arrived

The one artifact that does not care whether the cage Wi-Fi is up. Print before they walk in.

And it earns its keep the rest of the time: RMA paperwork, packing slips and shipping labels all have to be printed on site anyway.



The same card, print theme — white ground, one filled cell. Black backgrounds cost a fortune in ink.

What to actually change on Monday

01

Move remote hands off the phone and into a chat channel.

Bidirectional media, scrollbar, and a record you did not have to write.

02

Check connectivity before the window, and print a copy anyway.

Cage Wi-Fi is marginal and it is not yours to fix.

03

Never send a bare slot or bay number.

Index base, major axis, origin corner and bank are all unstated — and all variable.

04

Send a picture, and keep red for what does not come back.

Orientation, landmarks, slot name on the cell — and a separate color for parts that are only moving.

05

Identify the chassis by locator LED and serial, never a sticker.

Let the AI work out how to light that particular model.

06

Distill the manuals into a skill, then verify the geometry.

Figures mirror silently. Confirm against a live host or a photo, then pin it.

07

Match the format to the technician, and put a printer in the DC.

The best card is the one they will use. Paper also covers RMAs and shipping labels.

A picture is worth 1024 bytes. Give them a K.